COMPARATIVE TECHNICAL REPORT

Wildlife-Crime Monitoring Platforms

Coverage, taxonomic breadth, event traceability, visible routes, update cadence, data openness and researcher access

Independent evidence-based comparison | 8 November 2026

Executive summary

This assessment compares six resources: the Illegal Wildlife Trafficking Global Observatory (WCS Brasil), TRAFFIC's Wildlife Trade Portal, the Environmental Investigation Agency (EIA) Global Environmental Crime Tracker, FREELAND Brasil's Observatório do Tráfico de Fauna, the C4ADS Wildlife Seizure Dashboard, and ECO-SOLVE's Global Monitoring System (GI-TOC). The comparison is restricted to what an independent researcher can substantiate from public documentation and accessible public-facing resources. It does not rank institutional competence or the quality of undisclosed internal databases.

The strongest distinction is not "which platform is largest?" but what the platforms actually let a researcher verify, retrieve and reproduce. The WCS Brasil Observatory exposes a structured public event feed and associated route evidence on GitHub, with documented record fields and a recent commit history. TRAFFIC also provides incident-level data, visible trade routes and export functionality. EIA provides incident drill-throughs and route visualizations but requires a request for its full dataset. FREELAND Brasil publishes interactive regional summaries and directs users to a data-access form. C4ADS is strong on a deliberately restricted set of six wildlife categories and reports daily updates, although full raw-data accessibility is not established from public pages. ECO-SOLVE is strongest in systematic online-market monitoring, a different unit of analysis from seizures. [1–18]

No platform can be declared the most complete, the fastest, or the most scientifically reliable from the sources examined. Record totals are not normalized; update frequency is not consistently observable; and a public repository does not automatically confer legal permission to reuse its contents.

Key judgments

- For independently re-analysable seizure-event research, WCS Brasil and TRAFFIC offer especially relevant public data structures; EIA is also rich at the event level but requires a request for the full dataset.
- For documented routes on a map, WCS Brasil, TRAFFIC and EIA have clear public evidence; C4ADS displays country-level “top routes”, which is not the same as a verified event-level itinerary.
- For online advertisements, seller behaviour and market pressure, ECO-SOLVE has a specialised monitoring capability not directly matched by a seizure-event database.
- Daily monitoring, daily ingestion, daily publication and daily data freshness are four different claims; they must never be treated as synonyms.

1. Scope, method and rules of interpretation

Reviewed 8 November 2026 using official platform pages, operator methods/guides and the WCS Brasil public repository. No authenticated cross-platform benchmark was conducted; this is a documentation-led comparison with limited public-interface verification.

What is compared

- Coverage: geography, years and sampling scope; not presumed completeness of trafficking.
- Taxonomy: all groups versus prioritised groups; species-level availability and uncertainty.
- Event detail: date, location, taxon, quantity, mode, description, source and identifier.
- Routes: mapped paths, event links and observed versus intended versus inferred movements.
- Cadence: monitoring, ingestion, public release and measurable publication lag.
- Traceability/access: source evidence, downloads, reuse rights, version history and requests.

Evidence coding

Documented = stated by the operator or visible in public repository files. Partially evidenced = described but not tested at row/export level. Not established = insufficient public verification, not proof of absence. Access by request = provider states or enables a data enquiry. A “daily” claim is dated and kept distinct from demonstrated public-data latency.

Important comparability safeguards

Advertisements, seizures, investigations, prosecutions and routes are distinct units. Focused taxonomic datasets may be deeper than broad datasets. Counts differ in period, definition, aggregation and deduplication; they cannot be ranked directly. A seizure-location map does not establish an actual transportation route. [3, 5, 7, 9, 13–18]

2. Platforms included

WCS Brasil | Illegal Wildlife Trafficking Global Observatory

Global event-level OSINT repository; public canonical `cases.json` and `routes.json`, multilingual case fields and location/precision attributes; operational case additions evidenced in November 2026 GitHub

commits. Current audited event count, country/taxon totals and update-lag distribution were not recomputed for this report. [1, 2, 19]

TRAFFIC | Wildlife Trade Portal

Global wildlife-trade incidents and seizures, mapped routes, incident exploration and data export. The public homepage reports approximately 42,000 incidents and 8,900 trade routes; these are provider totals, not independently audited or commensurate with WCS's event count. TRAFFIC's 2020 documentation required registration, while current public materials advertise open access. [3–5]

EIA | Global Environmental Crime Tracker

A global environmental-crime resource covering selected wildlife taxa and other sectors. Provides individual incident drill-through, mapped seizure sites and trade routes, prosecution information, and a full dataset obtainable by request. Provider says data are regularly refreshed but does not define a daily public cadence. [6–9]

FREELAND Brasil | Observatório do Tráfico de Fauna

Media-derived seizures reported in Brazil, Argentina and Paraguay, beginning in 2018 in the published series. Public visualizations focus on item categories, animal groups, regions and quantities; new records are added when a new report is published. Underlying data are linked via user-type-specific access forms. [10–12]

C4ADS | Wildlife Seizure Dashboard

Global seizure analytics focused on six wildlife categories: elephant ivory, leopard, lion, pangolin, rhino horn and tiger. The public page reports 8,467 seizures in 124 jurisdictions since 2013. C4ADS's 2024 report describes daily updates, but public raw-row export and present-day daily release were not verified. [13–15]

GI-TOC | ECO-SOLVE Global Monitoring System

A distributed monitoring system for online wildlife advertisements, with country/region-specific species baskets and platforms, manual monitoring and AI-enabled discovery. Global reports contain detailed market analyses; this review could not verify public access to a complete row-level advertisement feed. [16–18]

3. Comparative matrix A — coverage, taxonomy and event-level detail

Criterion	WCS Brasil	TRAFFIC	EIA	FREELAND	C4ADS	ECO-SOLVE
Primary unit	Seizure / wildlife incident	Wildlife trade incident / seizure	Incident, seizure, arrest, prosecution	Reported seizure, aggregated for dashboard	Seizure in six categories	Online wildlife advertisement / listing
Geographic scope	Global intent; coverage shaped by documented source discovery [1]	Global; provider totals drawn from international records [3]	Global, with uneven monitoring effort and focal regions [6,7]	Brazil, Argentina, Paraguay [10]	Global; 124 jurisdictions shown [13]	Distributed national / regional monitoring hubs [16,17]
Taxonomic coverage	Cross-taxon case records; taxon name can be entered; completeness not audited [1]	Cross-taxon; filter by taxa, species, products [3,4]	Priority elephants, pangolins, rhinos, Asian cats, totoaba, plus incidental species [7]	Wildlife groups; species named where source permits [10,12]	Six categories only: ivory, leopard, lion, pangolin, rhino, tiger [13]	Hub-selected priority species baskets, across animals and derivatives [17]
Temporal coverage	Multiple event years; first/last date not independently audited [1]	Historic data; homepage examples from 2016 onward [3]	Varies by topic; Asian big-cat dashboard since 2000 [7]	2018 onward in published series [10]	2013–present, as labelled by dashboard [13]	Global report monitoring from April 2024 to April 2026 [17]
Individual event details	ID, event and publication dates, taxon, place, coordinates, mode, titles, narrative, source URL [1]	Incident details include species, products, location, route data [3–5]	Incident drill-through, unique ID, description and item rows [7]	Public page emphasises aggregated charts; record-level access via form [10,11]	Filters, summaries, countries and methods; row-level fields not fully audited [13,14]	Listing-level internal records described; public row-level feed not verified [16–18]
Quantity and product form	Structured counts, kinds and notes; weights not silently converted into counts [1]	Commodity counts and weights; user-defined metrics [3]	Per-item data; some estimates and imputations disclosed [7]	Counts / kilograms by item type and group [10]	Seizure-category trends; underlying quantity granularity not independently verified [13]	Listings, items and products analysed; not seizure quantities [17]

Source numbers in brackets refer to the reference list. “Not established” means not verified by the public evidence reviewed. Public datasets, operator claims and restricted internal capabilities are not interchangeable.

4. Comparative matrix B — spatial specificity and route evidence

Criterion	WCS Brasil	TRAFFIC	EIA	FREELAND	C4ADS	ECO-SOLVE
Transport / mode	Explicit modal field per case [1]	Transport methods and transit modes searchable [4]	Transport modes, concealment and multi-country routing [7]	Not established in public summary views [10]	Transport-sector filters and trends [14]	Not a consistent seizure-mode database; marketplace channel is central [17]
Geospatial precision	Lat/lon and precision metadata required in schema [1]	Map of incidents and trade routes; precise error metadata not audited [3]	Incident lat/lon mapping documented [7]	Regional choropleth/hotspot map [10]	Jurisdiction, city and country-profile views [13,14]	Market geography through monitoring hubs; event coordinates not comparable [17]
Visible routes on map	Yes: documented segments in shared MAP/GLOBE route feed [2]	Yes: interactive route maps; ~8.9k routes reported [3]	Yes: trade-route map, including incident drill-through [7]	Seizure-location map shown; route paths not established [10]	Top trafficking routes at country level; event-path mapping not verified [14]	Market flows discussed; public event-linked route map not verified [17]
Route provenance	Evidence, source URL, type, endpoints, direction, precision and event match [2]	Route-related incident fields; detailed source certainty needs record audit [3,5]	Origin/transit/outbound/destination fields; date imputation also documented [7]	Not established in public-facing regional dashboard [10]	Public route summaries; event-specific provenance not independently verified [14]	Not equivalent to transport trajectories of seizure events [17]
Source traceability	Source label and URL mandatory; multilingual explanatory context [1,2]	Record detail and public-source methodology; inspect each result for source links [4,5]	Public-source approach; details and descriptive source content [6,7]	Based on online news and agency releases [10]	Publicly reported seizures; direct link completeness not audited [13,14]	Evidence review and documentation internal; public ad URLs not verified [16–18]

Source numbers in brackets refer to the reference list. “Not established” means not verified by the public evidence reviewed. Public datasets, operator claims and restricted internal capabilities are not interchangeable.

5. Comparative matrix C — cadence, open data and access

Criterion	WCS Brasil	TRAFFIC	EIA	FREELAND	C4ADS	ECO-SOLVE
Daily monitoring / updates	Daily OSINT workflow; Nov 3–6 commits confirm frequent additions, not guaranteed daily releases [19]	No fixed daily release cadence verified [3–5]	“Regularly refreshed”; no fixed daily cadence verified [6]	News followed daily; dashboard additions coincide with reports [10]	2024 institutional report says dataset updated daily; 2026 latency untested [15]	Continuous monitoring; public dashboard refresh cadence not verified [17]
Public structured download	Direct public JSON feeds; manual parse feasible [1,2]	Export underlying data in portal [3]	Full dataset upon request; dashboard has incident drill-down [6–9]	Form-mediated data access [11]	Downloadable graphics/maps; raw-record bulk export not confirmed [14]	Complete ad-level export not established in public access test [16–18]
Must request data?	No request to retrieve public canonical JSON [1]	Self-registration historically required; no bespoke request proved necessary [4]	Yes for complete dataset / prosecution Excel [6–9]	Form(s) for governmental and other users [11]	Not for public dashboard; full underlying database may require enquiry [14]	Likely needed for individual ads / metadata; terms to negotiate [16–18]
Clear open-data licence?	Not identified in repository metadata; public access ≠ reuse permission [1]	Public access/export stated; reuse licence requires review [3]	Access free, full data on request; public licence not established [6]	Access terms depend on request form; licence not established [11]	Raw-data reuse terms not established [13,14]	Public reports; underlying individual dataset rights not established [16,17]
Version / change audit	Git commit history; validation rules and schema [1,19]	Portal methodology; complete public change-log not verified [3–5]	User guide documents methods; per-record version log not verified [7]	Series/edition-specific reporting; per-record log not verified [10]	Methodology public; per-record version log not verified [14,15]	Report series and hub approach; public row-level version audit not verified [17]

Source numbers in brackets refer to the reference list. “Not established” means not verified by the public evidence reviewed. Public datasets, operator claims and restricted internal capabilities are not interchangeable.

6. Interpretation of the results

6.1 Traceability is multidimensional

Direct access to a source URL is a meaningful advantage, but not the whole chain of evidence. A traceable record also requires a stable event identifier, explicit event/publication dates, revision history, documented coordinate precision, quantitative units, linked route evidence, and a distinction between “reported”, “observed”, “planned” and “inferred”. WCS Brasil’s published schema and route feed represent a relatively explicit design on these dimensions. TRAFFIC and EIA likewise supply event-level information and route visualizations, and EIA discloses important date-estimation rules [1–7].

6.2 Visible routes: not all maps answer the same question

Three platforms have documented route visualizations for individual or selected incidents: WCS Brasil, TRAFFIC and EIA. WCS Brasil explicitly stores segment-level geometry, route type, directional logic and narrative evidence. TRAFFIC provides large-scale route maps and searchable incident details. EIA’s user guide describes a route-map drill-through and country roles (origin, transit, outbound and destination). C4ADS highlights major routes in country profiles, but that should not be treated as equivalent to showing each verified itinerary as a line on the map. FREELAND Brasil’s public dashboard primarily maps seizure frequency; ECO-SOLVE’s ads-based market intelligence is not a shipment-route register. [2,3,7,10,14,17]

6.3 Taxonomic breadth and analytical depth involve trade-offs

Broad taxonomic coverage is valuable for cross-species comparison and early detection of neglected groups. But declaring broad coverage is not evidence of taxonomic completeness, correct species identification or consistent identification depth. TRAFFIC and WCS Brasil pursue wider taxonomic scopes than C4ADS’s six-category dashboard or EIA’s focal-species strategy. ECO-SOLVE uses hub-specific baskets, a rational approach for online monitoring in distinct markets. Benchmarking should report the proportion identified to species, genus and higher taxonomic levels, and the percentage of records with ambiguous names. [1,3,7,13,17]

6.4 Daily updates require an auditable service-level metric

The distinction among searches, ingestion, publication and source-event dates is critical. WCS Brasil’s GitHub history shows successive validated case additions on 3, 5 and 6 November 2026; this documents active maintenance, not continuous 24-hour data freshness. C4ADS explicitly stated in its 2024 annual report that its dataset was updated daily, but that historical statement is not a live measure of the 2026 data pipeline. FREELAND Brasil clearly states that its dashboard is updated when a report is issued, despite daily news monitoring. EIA describes regular refreshes without a daily promise; the public ECO-SOLVE release schedule does not establish the daily availability of individual listing data. [10,15,19]

6.5 Open data: access, download, rights and reuse

Open data should be evaluated across four distinct conditions: no discretionary access barrier; machine-readable downloads; sufficient metadata and documentation; and explicit permission for reuse, redistribution and derivative work. WCS Brasil’s canonical files meet the first two conditions visibly, with schema and version history supporting the third. However, a clearly stated reuse licence was not identified during this review. TRAFFIC publicly advertises data export, with registration mentioned in its 2020 launch

material; licence conditions must be checked separately. EIA offers the entire dataset on request, whereas FREELAND Brasil directs users to forms. In C4ADS and ECO-SOLVE, a public dashboard does not in itself prove bulk row-level data accessibility. [1,3,4,6,11,14,16]

7. What each platform is demonstrably well suited for

WCS Brasil Observatory

Cross-taxon incident investigation when researchers need primary-source links, event and publication dates, explicit route evidence, georeferencing and independently parsable files. Its main unverified dimensions are coverage completeness, systematic latency, user-facing quality assurance and formal reuse licensing.

TRAFFIC Wildlife Trade Portal

Large global incident queries, species/commodity comparisons, transport modes, route maps and exportable records. An appropriate principal external comparator for incident-level research; differences in event definitions and curation still require harmonization.

EIA Global Environmental Crime Tracker

Detailed analysis of priority-species enforcement, routes and prosecution outcomes, together with methodological disclosures. Useful for case-level queries, while full-dataset reuse depends on a request.

FREELAND Brasil

Regional trends and media-derived seizure indicators in Brazil, Argentina and Paraguay, with published periodic reports. Less suited to independently assembling case-level route evidence from its public summary interface alone.

C4ADS Wildlife Seizure Dashboard

Focused longitudinal comparisons across six frequently trafficked categories, geographic profiles and transport-related trends. A promising comparator for update cadence, but not a surrogate for a broad all-taxa monitoring system.

ECO-SOLVE GMS

Online marketplace surveillance, platform/seller behaviour, advertising trends and protected species offered for sale. Complementary rather than directly substitutable for seizure-event databases; public listing-level access remains unresolved.

8. Biases, limitations and open questions

- Seizure databases observe enforcement and disclosure, not the latent volume of illicit trade. Jurisdictions with stronger enforcement or more active media may appear over-represented.
- Online-listing data observe supply signals, not necessarily completed transactions or criminal offences. Some listings may be legal, fraudulent, duplicate or misclassified.
- Platform-reported totals were collected at different times and follow different definitions. Raw incident counts cannot be used to rank coverage or effectiveness without a harmonized sampling design.

- Routes are particularly vulnerable to overinterpretation. A claimed destination, transit point, intended journey and confirmed transport are not equivalent. A line on a map should disclose its evidence status.
- An absence of publicly verified features in this review must be treated as uncertainty, not a demonstrated absence of an internal capability.
- This analysis did not run controlled, logged test queries across all authenticated dashboards, reconstruct their downloadable tables, or independently sample the validity of underlying cases.

9. Reproducible benchmarking protocol for a future study

A rigorous next phase should use the same pre-registered test design for all six services, with two separate tracks: seizure-event systems and online-market systems. Do not assign a numerical winner until the raw observations and access restrictions have been logged.

Coverage test — Select a stratified reference set of officially documented events from several continents and years; test recall and false matches without treating absence as an automatic platform failure.

Taxonomy test — Use a pre-defined species basket (e.g., birds, reptiles, fish, mammals, invertebrates, parts and derivatives) and record species-level identification, synonyms and missing taxa.

Traceability test — Randomly sample 30 events per eligible platform; check dates, country, city, quantity, mode, source link, source availability and revision indicators.

Route test — For 20 incidents with external route evidence, record whether the route is visible, linked to an event, directed correctly and distinguishes completed from intended movement.

Freshness test — For 30 consecutive days, snapshot public feed timestamps and a reference stream of newly disclosed events; calculate median public-release lag and the share published within 1, 3 and 7 days.

Open-data test — Document access steps, registration, manual permission, full-row export, column dictionary, API/download reproducibility, version snapshots and licence terms.

User-experience test — Time identical scientific tasks (find an event, export all incidents of one species, reconstruct a route and reach the original source), with two independent testers.

Error and correction test — Submit a documented correction and record acknowledgement, time to amendment, changelog visibility and whether aggregate outputs update consistently.

Publish the search protocol, benchmark query strings, test log, code, sampled public IDs and uncertainty rules wherever licence, security and personal-data restrictions permit. Treat restricted systems separately from public-access claims.

10. Conclusions

The WCS Brasil Observatory is notably useful to independent researchers because structured case data, source links and route evidence are accessible without a negotiated data-sharing arrangement, and because its public repository makes revisions observable. This is a meaningful operational design choice, not proof of superior coverage, accuracy or daily completeness.

TRAFFIC remains the closest public peer for wide-ranging incident-level studies and deserves equal methodological attention because it also exposes incidents, routes and exports. EIA offers meaningful traceability and incident detail, although its full underlying data are request-mediated. FREELAND Brasil is an important regional comparator with report-cycle updates and a data-request workflow. C4ADS

demonstrates focused taxonomic analytics and makes a specific historical daily-update claim. ECO-SOLVE is stronger as a specialised online-advertisement monitoring network, but its public data granularity cannot be equated with its internal capabilities.

The most defensible comparative claim is therefore conditional: for research prioritising openly retrievable, source-linked, geo-referenced wildlife seizure events and route evidence, WCS Brasil offers an unusually accessible combination of features; whether it surpasses TRAFFIC or EIA in completeness, update speed or accuracy remains an empirical question. Publishing a licence, maintaining a machine-readable change log and auditing record-level publication lag would strengthen this proposition considerably.

References and verification links

[1] WCS Brasil / GitHub. Wildlife-trafficking-data repository, case schema and README. Source link

[2] WCS Brasil / GitHub. Canonical documented route feed (routes.json). Source link

[3] TRAFFIC. Wildlife Trade Portal, live homepage, maps, metrics and export. Source link

[4] TRAFFIC (2020). New tool to track wildlife trade; user registration and incident exports. Source link

[5] TRAFFIC. Portal data upload and report-error workflow. Source link

[6] EIA. Global Environmental Crime Tracker, access and full-dataset request. Source link

[7] EIA (updated 2023). Global Environmental Crime Tracker user guide, especially pp. 3, 10, 18–19. Source link

[8] EIA. Global Environmental Crime Tracker data enquiry form. Source link

[9] EIA. Description of wildlife focus groups and regular database refreshes. Source link

[10] FREELAND Brasil. Interactive Observatório do Tráfico de Fauna: coverage and update-by-report rule. Source link

[11] FREELAND Brasil. Access-to-data forms for governmental and other users. Source link

[12] FREELAND Brasil. Observatory description and public update indicator (Portuguese). Source link

[13] C4ADS. Wildlife Seizure Dashboard homepage, reported counts and six taxonomic categories. Source link

[14] C4ADS. Dashboard About, methodology and country/transport features. Source link

[15] C4ADS (2024 Annual Report, published 2025), pp. 18–19: “updated daily” statement. Source link

[16] GI-TOC. ECO-SOLVE initiative and monitoring mission. Source link

[17] GI-TOC (7 July 2026). ECO-SOLVE sixth Global Trend Report and hub-specific species baskets. Source link

[18] ECO-SOLVE. Public Global Monitoring System dashboard (dynamic interface; raw export unverified). Source link

[19] WCS Brasil / GitHub. Main-branch commit history; November 2026 additions. Source link

Prepared 8 November 2026. URLs identify publicly available provider documentation; link availability, datasets, interface details and access policies may change. The report is intended as a documented comparison, not an endorsement.